Shahjalal University of Science & Technology
Department of Computer Science and Engineering
4th Year 1st Semester Final Examination 2023
Course: CSE 461 (Introduction to Computer Security and Forensics)
Credits: 3 Marks: 100 Time: 3 Hours

Group A
(Answer ALL the questions)

1. **Answer any FIVE** 5x2=10
   - a) What do you understand by confidentiality? Mention four tools used to achieve it.
   - b) What do you mean by biometric? Mention four candidates for biometric IDs.
   - c) Which attack could be posed on One Time Pads encryption method?
   - d) What are the steps of worm/malware development?
   - e) "Choosing switch over hub is a good decision from a security aspect"- do you agree? Why or why not?
   - f) How ARP spoofing can be prevented? Give two reasons at least.
   - g) N users in a centralized network will communicate **only** with the server using symmetric key encryption. How many different keys are needed?
   - h) Differentiate between second image resistance and collision resistance in Cryptographic Hash function.

2. **Answer any FOUR** 4x5=20
   - a) Discuss ten security principles in brief.
   - b) Explain Shannon's confusion and diffusion properties. Also identify correct association of AES methods with them.
   - c) What is Computer Forensics? Describe Forensic Paradigm.
   - d) What is buffer overflow? How can you conduct buffer overflow attack? Explain with stack layouts.
   - e) Discuss five different types of malwares in brief.
   - f) In RSA signature, write down the signature and verification steps and provide the correctness of verification.

2x10=20

c) What is Computer Forensics? Describe Forensic Paradigm.

d) What is buffer overflow? How can you conduct buffer overflow attack? Explain with stack layouts.

e) Discuss five different types of malwares in brief.

f) In RSA signature, write down the signature and verification steps and provide the correctness of verification.

3. **Answer any TWO** 2x10=20

a) "If approximately 58 people are present in a room, there is 99% chance that at least two of them has same birthday"- prove this birthday paradox statement in the context of collision of hash values of different messages. 10

b) Assume that the following text block for AES encryption after the SubBytes step (contains only hexadecimal values; each bigram represents 1 byte of text) 2+5+3

{10, 07,1a,13,0e,23,17,03,2e,01,02,17,0d,0f,1a,09}

and the round key is given as

{01,01,01,01,01,01,01,01,01,01,01,01,01,01,01,01}

you may assume the first row of mix const matx[02, 03, 01, 01]

i. Apply Shift Row step
ii. Apply Mix Column step
iii. Apply Add Round Key step
- to find the encrypted block

c) Suppose Alice wants to send a message M = 8 to Bob using the Elgamal cryptosystem. Bob has a prime number, p = 59, and a primitive root in Z_p, g = 2, and y = 25. 1+4+5

i. What is the public key pair that Alice will use to encrypt the message M?
ii. Show the result of encryption using the public key when k = 4.
iii. Now, Bob wants to decrypt the message sent by Alice using his private key x. Show the correctness of the decryption process.

Group B

(Answer ALL the questions)

4. **Answer any FIVE** 5x2=10

   a) Differentiate between authorization and authentication.

   b) Mention six possible evidence's sources of forensic analysis.

   c) Calculate the search space if a system used 2-byte salt and the attacker had a 1 million-word dictionary for attacking salted passwords?

   d) Differentiate between polymorphic virus and metamorphic virus.

   e) What is meant by the 10-character access control notation **drwxr- - - -x**?

   f) Differentiate between rule based and statistical intrusion detection system.

   g) Mention two different attacks based on DHCP in brief.

   h) Differentiate between HTTP GET and POST request.

5. **Answer any FOUR** 4x5=20

   a) What is digital signature? Write down its properties.

   b) How "Smurf attack" is implemented? Explain with necessary figure/flow diagram.

   c) Define adware and spyware with necessary flow diagram.

   d) How is SSH connection established between a client and a server?

   e) Differentiate between Static and Dynamic Analysis of a suspicious file in a system.

   f) How does DNSSEC prevent DNS cache poisoning? Explain with a figure.

6. **Answer any TWO** 2x10=20

   a) In RSA, two prime numbers are picked, p = 11, q = 3, and e = 7. 6+2+2

      i. Find the private key d.

      ii. If message M = 31. Find the cipher C.

      iii. Find the decipher text after decryption and compare it with plaintext.

   b) i. Explain how **HTTPS/TLS** protocol works to establish a session to transfer data with proper diagrams. 5+5

      ii. How session hijacking attack can be imposed? What measures can be taken to mitigate this attack? – answer briefly.

   c) i. Describe URL obfuscation attack with example. 5+5

      ii. Describe SQL injection attack with example.

$2^3 \times 2^3 \times 2^3 \times 2^3 \times 2^3 \times 2 \times 3$

$(e, n) = (7, 33)$ $C = ⑥$

4^2 3^{23} $31^{33} \bmod 7$

$4^3 \times 4^3 \times$

$n = p * q = 33$

3

$C = P^e \bmod n$

$1^{23} \bmod 33$

$\phi(n) = (p-1) * (q-1) = 20$

$P = C^d \bmod n$

identification

collection

analysis examination

reporting

$(4^{16} \times 4^4 \times 4^2 \times 4) \bmod 33$

09

31

$d = e^{-1} \bmod \phi(n)$

3

31

$\phi(n) d$ [illegible] $+1$

1984

1980